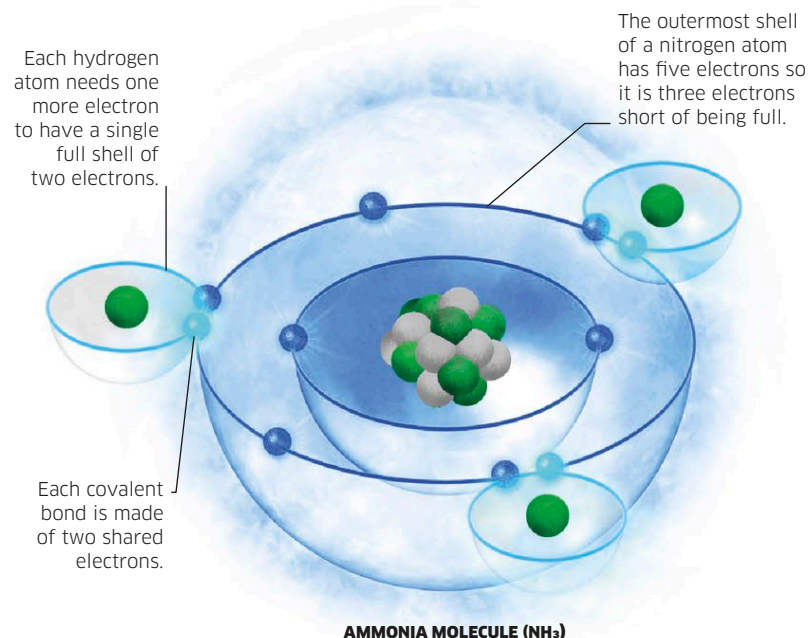


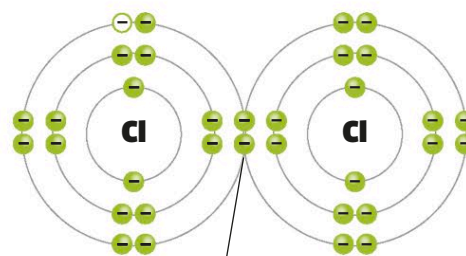
Covalent bonding

Another way atoms can attain full outer electron shells is by sharing electrons in a covalent bond. A molecule is a group of atoms held together by covalent bonds (see pp.14-15). Some elements exist as molecules formed by pairs of atoms, for example chlorine, oxygen, and nitrogen. Covalent bonds can be single, double, or triple bonds.



Ammonia molecule

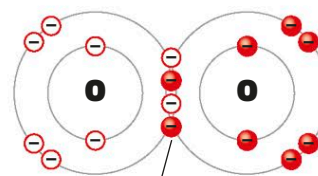
A molecule of the compound ammonia (NH₃) is made of atoms of nitrogen (N) and hydrogen (H). The shell closest to the nucleus of an atom can hold only two electrons. Hydrogen and helium are the only elements with just one shell.



One pair of shared electrons completes each chlorine atom's outer shell.

Single bond

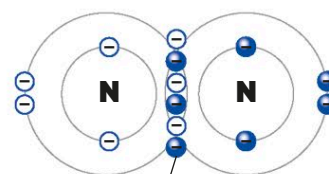
Some pairs of atoms share only one electron each, forming a single bond.



Oxygen atoms share two electrons each to fill their outer shell.

Double bond

Sometimes, pairs of atoms share two electrons each, forming a double bond.



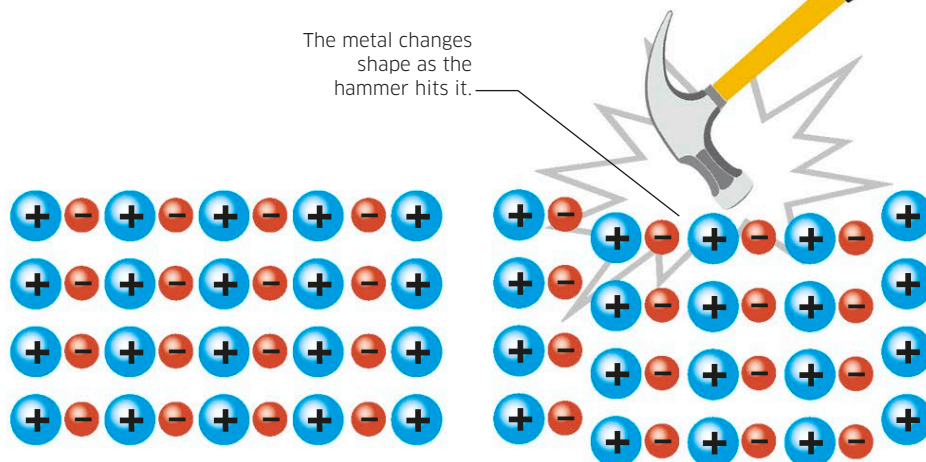
Before bonding, a nitrogen atom's outer shell is three electrons short, so it forms a triple bond.

Triple bond

In some pairs of atoms, three electrons are shared, forming a triple bond.

Metallic bonding

In a metal, the atoms are held in place within a "sea" of electrons. The atoms form a regular pattern—a crystal. Although the electrons hold the atoms in place, they are free of their atoms, and can move freely throughout the crystalline metal. This is why metals are good conductors of electricity and heat.

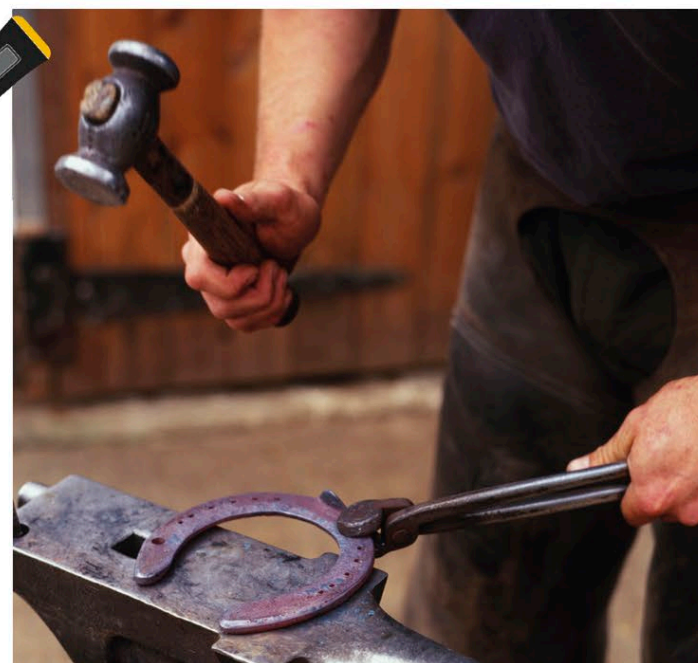


Conducting heat and electricity

An electric current is a flow of electric charge. In a metal, negatively charged electrons can move freely, so electric current can flow through them. The mobile electrons are also good at transferring heat within a metal.

Malleable metals

Metal atoms are held in place by metallic bonding, but are able to move a little within the "sea" of electrons. This is why metals are malleable (change shape when beaten with a hammer) and ductile (can be drawn into a wire).



Getting into shape

With some heat and a hammer, metals can be shaped into anything from delicate jewelry to sturdier objects, such as this horseshoe. Horseshoes used to be made of iron, but these days metal alloys such as steel (see p.63) are more common.